National Institute of Technology Calicut
Department of Computer Science and Engineering

Winter Semester 2025-26
CS4086E: Systems Programming Lab
Practice-Lab – 01 (16, Jan. 2026)

Beginners

1. Read an integer N as input from the user and print all the prime numbers in the range 2 to N.

    Test Cases:

    1. Input: 10 Output: 2, 3, 5, 7
    2. Input: 50 Output: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47

2. Given an array, find the product of all non-zero elements in the array Test Cases:

    1. Input: [2, 0, 0, 1, -5, 7] Output: -70
    2. Input: [0, 0] Output: 0

3. Write a shell script that accepts a filename as a command-line argument and prints the total number of lines in the file.

    Input: ./linecount.sh data.txt
    Output: 5

4. Write a shell script to read an array of integers and print the largest element.

    Input: Enter numbers: 10 25 7 40 18
    Output: Largest element: 40

5. Write a shell script to print all even numbers between 1 and 50.

    Input: ./even.sh
    Output: 2 4 6 ... 50

6. Write a shell script that takes a string as input and prints its reverse.

    Input: OperatingSystem
    Output: metsySgnitarepO

7. Write a shell script that reads a string and prints it in uppercase.

    Input: helloUnix
    Output: HELLOUNIX

9. Write a script that takes an array of numbers and identifies if any of them are ”Perfect Numbers” (where the sum of divisors equals the number).

    Input: 6 10 28
    Output: 6 is Perfect, 10 is Not Perfect, 28 is Perfect

Intermediate

1. Read an integer N and print all the numbers less than N and relatively prime to N. (Note: An integer *a* is relatively prime to *N* if gcd(a, N) = 1)
   Input: 7 Output: 1,2,3,4,5,6
   Input: 10 Output: 1, 3, 7, 9

2. Write a shell script to count the number of files and directories in the current working directory separately.
   Input: ./count.sh
   Output: Number of files: 6
   Number of directories: 3

3. Write a shell script to check whether a given string is a palindrome.

    Input: Enter a string: ABBA
    Output: Palindrome

4. Find Largest and Smallest Element in an Array: Write a shell script that accepts an array of integers from the user and displays the largest and smallest elements.

    Input : 12 5 89 3 45
    Output : Largest: 89 Smallest: 3

5. Read two arrays and print the common elements.

    Input:
    Array1: 1 2 3 4
    Array2: 3 4 5 6

    Output: 3 4

6. Given an array of integers, count how many elements are positive (>0).

    Input: [2, -1, 0, 5, -3, 7] Output: 3
    Input: [0, -2, -5] Output: 0

7. Write a script that checks the size of a directory provided as an argument. If the size exceeds 100KB, print a warning; otherwise, print "Size OK".

Input: /home/user
Output: Warning: Directory exceeds 100KB

9. Problem Statement: Write a script that accepts exactly three command-line arguments representing sides. Validate that arguments are integers, check if they form a triangle, and verify if they follow the Pythagoras property.

Input : 3 4 5
Output : It is a Triangle and follows Pythagoras property.

Advanced

1. Write a shell program that accepts a filename and a word as command-line arguments.

The program should:

1. Print the line numbers where the given word occurs in the file.
2. Print the total number of occurrences of the word in the file.
3. Handle error cases like file not existing, filename or word not provided as command line arguments.

Assume:

- Matching should be case-sensitive.
- The word may appear multiple times in the same line.

Input file: sample.txt

Linux is an operating system
Linux is open source
I like Linux
Linux Linux everywhere

./wordcount.sh sample.txt Linux

Output:
Word found at line numbers : 1 2 3 4
Total number of occurrences : 5

2. Write a shell script to replace all occurrences of a given word with another word in a file and save the output to a new file.

    Input : ./replace.sh input.txt file document output.txt

    Output (output.txt):

        This is a simple text document.
        This document is for testing.

3. Find Second Largest Prime Number in an Array: Write a shell script that takes an array of integers as input and finds the second largest prime number in the array.

    Input : 3 5 7 11 4 10 13

    Output : 11

4. Write a shell script that:

- Accepts a filename as a command-line argument
- Prints each word and its frequency
- Output should be sorted in descending order of frequency

    Input (file.txt): linux is linux and linux is powerful

    Output:

        linux 3
        is 2
        and 1
        powerful 1

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